Where Vision–Language Models Fail at 3D Space: A VSI-Bench Evaluation of Three Open-Weight VLMs

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1 Setup and protocol

We evaluate three open-weight VLMs on VSI-Bench (Yang et al., 2025), 1316 questions over 288 egocentric scenes from ScanNet and ScanNet++. Every model receives the same input: 32 frames sampled at $\mathrm{round}(\mathrm{linspace}(0, N-1, 32))$, long side capped at 448 px, greedy decoding with 48 new tokens, and prompts copied verbatim from the official `vsibench_doc_to_text`. Scores use the official MRA implementation; a lenient unit-aware parser is reported alongside where the two disagree.

Protocol. Qwen3-VL-8B-Instruct, Gemma-4-12B-it and Molmo2-8B (all Apache-2.0, run locally on one H100; torch 2.14.1+cu126, transformers 5.22.1 / 4.57.1, driver 580.178.04). Tasks on their full sets: `route_planning` (Cat. 1, ACC, n=194), `object_abs_distance` (Cat. 2, MRA, n=834), `room_size_estimation` (Cat. 2, MRA, n=288). Official ACC/MRA copied verbatim; 95% CI by bootstrapping over scenes. A *blind* arm repeats the numeric tasks with no frames.

2 Results

Table 1 reports every task on its full set. Qwen3-VL leads on all three; Gemma-4's strict numeric scores are near zero because its answers are prose that the official first-token parser discards, which the parenthesised lenient scores expose.

Table 1: Main results (strict scorer, 95% CI over scenes). Parenthesised values show the lenient parser where it differs by more than 15 points.

Model	**Route planning** ACC n=194	**Absolute distance** MRA n=834	**Room size** MRA n=288
Qwen3-VL-8B	35.0 [28, 42]	47.9 [45, 51]	57.1 [53, 61]
blind	–	30.6	32.9
Gemma-4-12B	30.4 [24, 37]	0.3 [0, 1] (22.8)	0.0 [0, 0]
blind	–	0.0	0.0
Molmo2-8B	29.4 [23, 36]	0.0 [0, 0] (16.2)	11.5 [8, 14]
blind	–	0.1	32.0

The blind control quantifies how much of each score is prior recall. Qwen3-VL retains 30.6 and 32.9 MRA with no image; Gemma-4 retains nothing; Molmo2-8B scores *higher* blind on room size (32.0 vs 11.5) because a constant guess is a good hedge against a ground truth centred near 20 m^2.

Numeric diagnostics. A model that measures should track the ground truth ($r > 0$) with comparable spread; $r \approx 0$ with $\sigma_{\hat{y}}/\sigma_y \ll 1$ is prior anchoring.

Model	Task	r	$\sigma_{\hat{y}}/\sigma_y$	Parse fail.	Med. rel. err
Gemma-4-12B	Abs. dist.	0.2341	0.6624	96.8	54%
Gemma-4-12B	Room size	–	–	100.0	–
Molmo2-8B	Abs. dist.	0.3332	2.9955	99.9	100%
Molmo2-8B	Room size	0.7425	3.2377	38.2	71%
Qwen3-VL-8B	Abs. dist.	0.7143	0.6487	0.0	28%
Qwen3-VL-8B	Room size	0.8136	0.5758	0.0	21%

Preprint.

3 Failure analysis

The five panels below are representative failures, each showing the frames the model saw with their source indices, the question, the ground truth and the prediction. Table 2 groups 2789 failing questions into four categories.

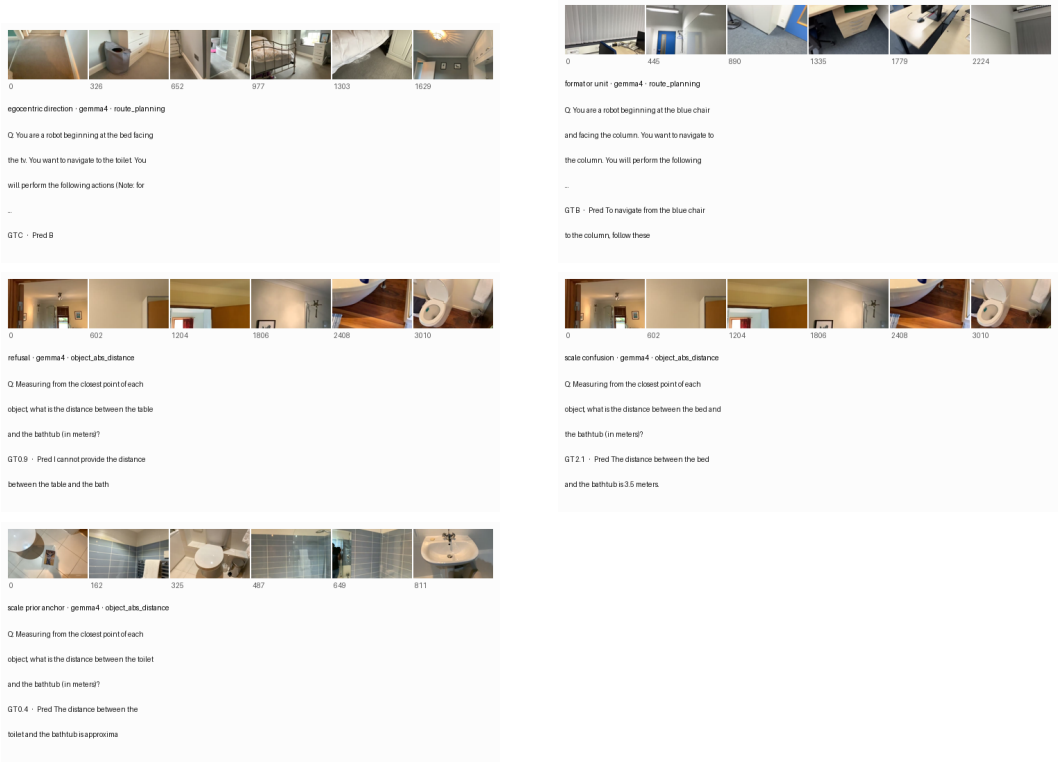


Table 2: Failure taxonomy. Near misses inside 30% relative error are excluded, since MRA already gives them partial credit.

Class	n	Mechanism
Scale confusion	1824	wrong magnitude; a prior-pinned subset sits near the task median
Egocentric direction	378	wrong turn sequence; B/C over-predicted, A/D under-used
Refusal / abstention	355	no estimate produced
Answer format	232	right in substance, lost to the first-token parser

4 Discussion

Qwen3-VL-8B is the only model of the three that measures rather than recites: 47.9 MRA on absolute distance and 57.1 on room size against 35.0 route-planning accuracy. Removing the video costs it 17.3 and 24.2 MRA points with non-overlapping intervals, yet a third of its distance score survives with no image at all, so part of any VLM's score here is prior recall in the benchmark as much as perception. Gemma-4-12B returns no room-size estimate on any of the 288 questions and needs the lenient parser to reach 22.8 on distances, so its 0.3 strict figure measures answer format, not spatial ability. Molmo2-8B tracks room size (r=0.74) but overshoots threefold, and its blind arm — a constant 20 m^2 — sits within a point of the best constant this ground truth allows, so MRA's thresholds reward hedging; route planning, where all three converge at 29–35%, is dominated by a letter prior rather than by failed turns. The largest remaining gain is in the input, not the model: 32 frames at 448 px cannot carry a room's metric scale, and asking the model to verify a proposed depth or floor plan would test spatial reasoning instead of recall. The limitation is structural — no frame budget repairs a benchmark whose numeric answers can be scored above chance by a constant.

References

J. Yang et al. Thinking in Space: How Multimodal LLMs See, Remember and Recall Spaces. CVPR 2025. arXiv:2412.14171.
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Z. Fan et al. VLM-3R: Vision-Language Models Augmented with Instruction-Aligned 3D Reconstruction. CVPR 2026.